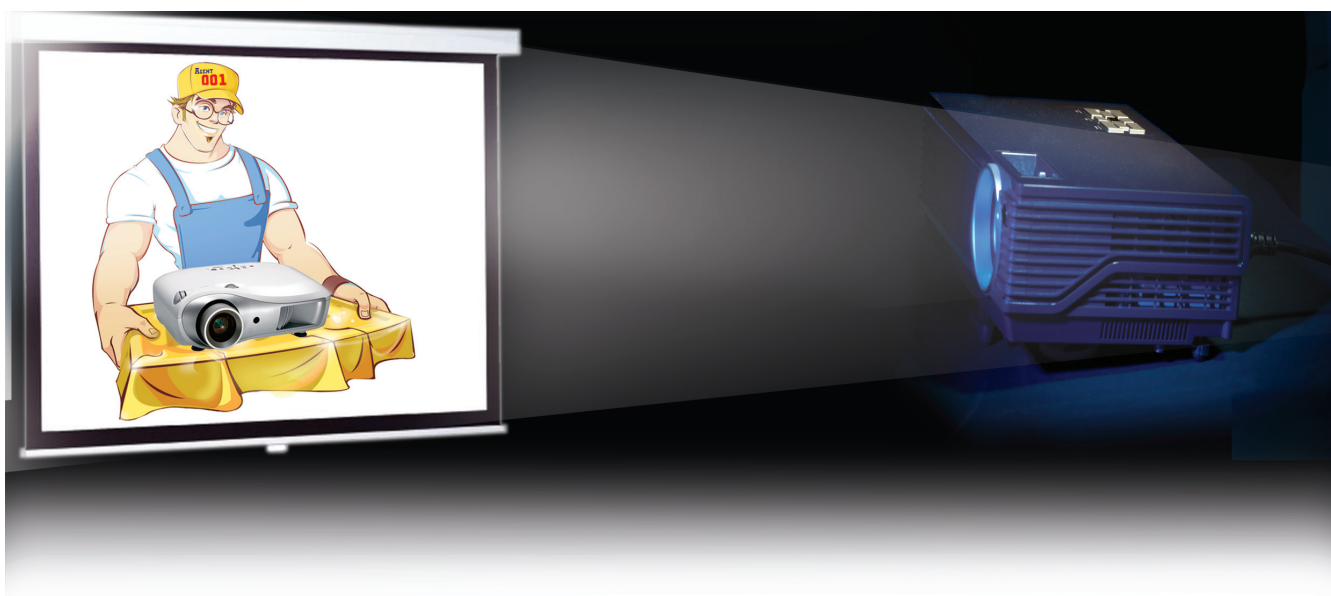


Projectors

110 inches of fun. Size does matter...buy one of these



With all the hype surrounding large flat screen panels most people tend to forget that besides LCD and plasma technologies there is another (often cheaper) way to get a huge display area. But projectors aren't common place. This is because until recently, projector manufacturers didn't want their products to become common place. Projectors enjoyed a steady demand, and those who wanted a projector, bought a projector. This was a niche category and vendors liked it that way. Prices were also suitably high. After all, there were no alternatives. Then came along the large LCDs and plasmas and although these aren't a direct threat to the projector, they did steal some of the limelight. We reviewed a couple of sub-Rs 50,000 projectors recently, and were quite pleased to report that display quality was pretty decent for the price.

Another reason for projectors remaining a niche market is their decidedly business-oriented origins. This is another factor which has changed considerably. These devices are becoming more home-oriented with support for HD resolutions and HDMI connectivity, amongst other goodies. A business projector rarely needed a close to perfect contrast

ratio or colours, as their primary applications were presentations in PowerPoint and Excel, along with HTML Web pages. It's shocking and welcomingly so, then, that manufacturers have reduced prices while improving specifications and adding support for HD resolutions like 720p (1280 x 720 pixels). The menu systems of projectors are also becoming simpler, so that anyone, even your 75 year old grandmother, can use it with the least fuss.

When I mentioned that projectors do not directly compete with flat panels, I was talking about a price competition. Price for you should be subject to two factors — screen size and quality. A 32-inch LCD TV may cost you around Rs 45,000. A projector with the same visual quality may cost you Rs 75,000, with an extra Rs 15,000 for a good screen (yes, projectors need those!). But such a projector can give you a display size of close to 100 inches. A 100-inch plasma panel would cost you at least four times as much as the projector, while LCDs of this size are extremely rare. Therefore, projectors compete very favourably in terms of price, and offer an unmatched cost per inch of display. Of course, 1080p projec-

tors are rare, while for flat panels this is commonplace. Most HD resolution projectors support up to 720p natively, else XGA (1024 x 768) is a *de facto* resolution. From my experience the additional resolution hardly detracts from the overall experience of movie watching, for example. Although the difference between a native 1080p and 720p projector is quite noticeable (with 1080p content), you will not complain with the 720p projector unless both are working side by side. Even XGA projectors do HD resolutions well, although the black strips on the top and bottom of the screen will detract from the viewing experience. The fact is that projectors do non-native resolutions very well, better than most LCDs and plasmas out there. This is also of interest to gamers, who may be running their games at non-native resolutions. From personal experience, I can tell you that gaming on a projector is much prettier than gaming on a 1366 x 768 pixel LCD or plasma display. The main attraction of a projector remains its huge display size — a true cinematic experience where size really matters!

There are mainly four types of projector depending on the technol-

ogy used. The first two types, CRT and LCOS, are prohibitively expensive, and quite frankly I've never come across either of these two types in the market. The other two types, i.e. LCD and DLP projectors, are fairly common. In general, DLPs are smaller and lighter than LCD projectors, although they do tend to heat up a bit more. I also feel that DLPs offer better contrast ratios, although some of the costlier LCD projectors are every bit as good. DLP projectors are, in general, better than their LCD counterparts. Entry level LCD projectors are priced at the Rs 35,000 mark or thereabouts whereas a cheap DLP projector will be at least Rs 45,000.

Business users will note that there is no distinct line between home and business projectors. A few differences may exist in the form of connectivity; obviously business users will need a wider array of display connects and wireless connectivity along with RS232 interface support, and even an RJ45 connect. However, the capabilities of the projector remain hardly unchanged. Obviously, the higher the resolution the better the projector for both business and home usage, but features such as native 720p support won't count for much for a corporate projector. HDMI and DVI connects are certainly useful for both segments of users.

When it comes to shopping for a projector you need to be a little discerning because you won't find them lying around in malls or electronics superstores. These are sold on demand, and very few vendors other than manufacturer retail outlets will even stock ready pieces. There are certain online stores which may ship a projector to your doorstep. The importance of knowing as much as possible about your projector is very important when you have to shop on specifications alone.

In my opinion, for home entertainment and such, the contrast ratio is of utmost importance. Look for the Static Contrast Ratio (SCR) value. This should be at least 1,500:1. This gives rich and vivid blacks and good intensity variation. The brightness level of the lamp is also important, particularly in case your room isn't totally dark; this is the case in most homes, where ambient light can spoil the overall experience.

Look for a brightness level of at least 2,500 lumens. For an absolutely dark room a brightness of 2,000 lumens is enough.

The next thing to look at is the distance between the projector and your screen / wall or whatever medium the image is being projected on to. This determines whether you need a short throw lens, or a long throw one. Projector lenses are interchangeable. Short throw lenses are good at projecting larger images over short distances, but they lose intensity if the screen is too far away. Long throw lenses project smaller images on a close-by screen. This is why they are used when your room is longer. They can also maintain a higher intensity beam for a longer distance.

The choice of screen is your next decision. This is a neglected part of most setups, and some people actually choose to ignore the screen altogether and opt for projecting an image directly on their living room wall. While this may be okay for the odd movie or game, if you are any kind of cinema buff you will want to spend at least Rs 15,000 on a good screen, and many experts believe that 25 per cent of your budget for the projector should be allocated to the screen itself. Brands like Pioneer and Zoher and Co are Indian brands that have been in the market for awhile. They have some decent offerings around the Rs 20,000 mark. There are cheaper options available, and I've seen generic Chinese brands available for as low as Rs 4,000 — these are also good, although the quality of materials may not be as good and the (short) longevity of the screen may be a problem. However, at the price of a good branded screen you can actually buy four of these — food for thought. Some of the higher-end screens are motorised — meaning that rolling them up is no longer a headache, but you can expect to pay more for such screens — in the region of Rs 4,000 to 10,000 more.

If you happen to get a demo of a projector that you want to buy, check on the working of the remote control unit. Some units do not work from certain angles — very annoying.

Among the brands, NEC comes highly recommended by me. Most of its models are priced within the

Rs 65,000 to Rs 1,20,000 range. If you can swallow this, there's a really sweet viewing experience in store for you that is rarely matched by similarly priced competing offerings. NEC's VT595 is a very decent entry to the world of projection viewing. At Rs 54,000 it's equally suitable to a boardroom and a living room. It includes both D-Sub and DVI connects — a plus for those intending to use it with a notebook for presentations, or an HTPC. It misses out on wireless and RJ45 connects. BenQ has a couple of decent entry-level projectors in the MP620c and the MP612c priced at Rs 44,000 and Rs 40,000 respectively. Movie buffs may want a little more visual quality and contrast, but these are some of the cheapest projectors with acceptable levels of quality around, and for the price I find the MP620c in particular quite good.

For me, one of the best deals ever is the Sharp XR-30X, which is a crisp XGA resolution supporting unit. We tested this piece earlier, and found it to be excellent in terms of contrast and colour rendition. The fact that it does equally well in both business and multimedia tests mean it's perfectly suited to both, although WiFi and RJ45 are missing. At a mere Rs 49,000 this projector puts many costlier ones to shame, proving that value and performance can coexist peacefully.

If you want HDMI connectivity in a superb multimedia projector, look at NEC's VT800. Perfect for those looking to spend near a lakh for something upmarket. It's got near perfect colour, clarity, brightness and contrast, and will really make HD and Blu-Ray content come to life. The LT380 (also from NEC) is a little better, but costlier, at Rs 1,24,000 and offers marginally better performance, and the price tag isn't worth the small performance premium. If you're a business user interested in presentations and graphs and the like, both these projectors are probably overkill — the Sharp XR-30X is a much better option. If you want something very close to the performance of the NEC VT800 at a much cheaper price tag, then look at the Sony VPL-EX5 (Rs 54,990). It's much cheaper, and performs nearly as well.